CREDIT RISK BEHIND THE SMILE

UNIVERSITY OF OXFORD

Wen Su

wen.su@maths.ox.ac.uk

Motivation

- Assessing corporate credit risk is essential for valuation and risk management. Credit ratings and credit default swap (CDS) spreads provide useful information, but the number of firms with CDS or rating is small.
- Equity option markets may contain information about corporate credit risk [Cremers et al., 2008, Carr and Wu, 2011].

	U.S. firms coverage
S&P ratings	$\approx$ 700 U.S. firms
CDS	$\approx$ 600 U.S. firms
Equity options	$\approx$ 3700 U.S. firms

Theoretical Framework

Asset Value, Default and Equity Vaule

- Let A_t denote firm asset value and V_t its instantaneous return variance:

$$\begin{aligned} \frac{\mathrm{d}A_t}{A_t} &= (r_t - q_t)\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_t, \\ \mathrm{d}V_t &= \mu(V_t)\,\mathrm{d}t + \xi(V_t)\{\rho\,\mathrm{d}W_t + \sqrt{1-\rho^2}\,\mathrm{d}W_t^{\perp}\}. \end{aligned} \tag{1}$$

- Default occurs when asset value reaches a deterministic threshold $D_u > 0$. Let $X_u = \log(A_u/D_u)$, we have $\tau_D = \inf\{u \geq t : X_u \leq 0\}$.
- Equity has value $S_t = s(t, X_t, V_t)$ before default and zero thereafter. For a equity maturity T_E, a common choice of s can be

$$s(t,x,v) = \mathbb{E}^{\mathbb{Q}}_{t,x,v}\left[e^{-\int_t^{T_E}(r_u-q_u)\,\mathrm{d}u}(A_{T_E} - D_{T_E})^+ \mathbf{1}_{\{\tau_D > T_E\}}\right]. \tag{2}$$

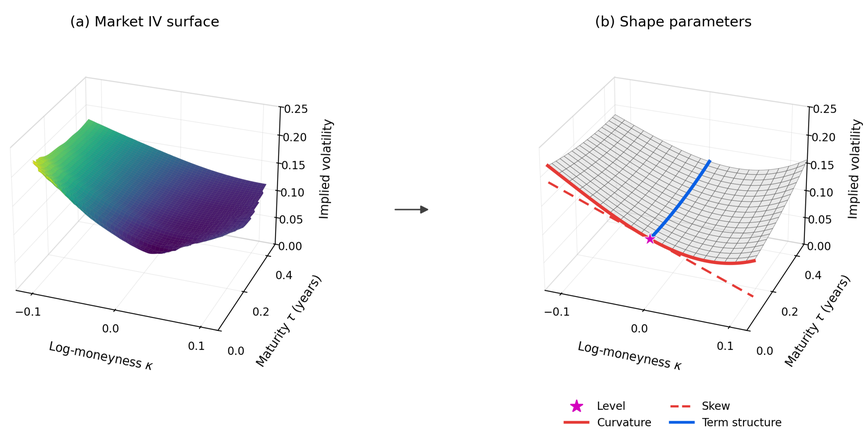
Equity Implied Volatility Expansion

Theorem 1 *Under some conditions, there are unique coefficient functions θ_{jm}, s.t.,*

$$\sigma_{\mathrm{IV}}(t,x,v;\kappa,\tau) = \sum_{j+2m\leq N} \theta_{jm}(t,x,v)\kappa^j\tau^m + O(\tau^{(N+1)/2}). \tag{3}$$

- A simple $N = 2$ example already has $R^2 \approx 0.93$ across firms:

$$\sigma_{\mathrm{IV}}(\kappa,\tau) = \underbrace{\theta_{00}}_{\text{spot vol}} + \underbrace{\theta_{10}}_{\text{slope}}\kappa + \underbrace{\theta_{20}}_{\text{curvature}}\kappa^2 + \underbrace{\theta_{01}}_{\text{term structure}}\tau + O(\tau^{3/2}). \tag{4}$$

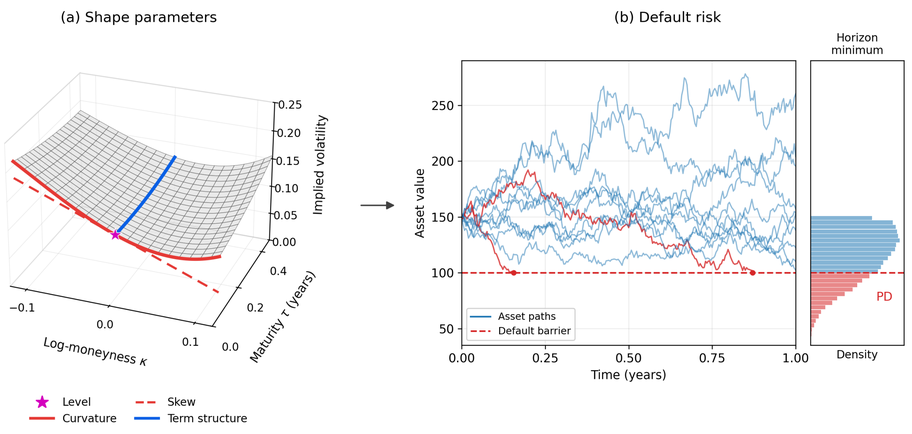


Default Probability Identification

- Default probability: $\mathrm{PD}_H(M) := \mathbb{Q}^M_{t,x_M,v_M}(\tau_D \leq t + H)$.
- $d_N(\boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2)$: distance between two IV shape parameter vectors.

Theorem 2 *Under some conditions, for every $H > 0$ and $0 < \eta < 1$, we have*

$$|\mathrm{PD}_H(M_1) - \mathrm{PD}_H(M_2)| \leq C_\eta\left(d_N(\boldsymbol{\Theta}_1,\boldsymbol{\Theta}_2)^\eta + e^{-c_\eta[\log\log(N+3)]^2}\right). \tag{5}$$



- Theorem 2 indicates that the IV coefficients can infer the default probability. But the form of this mapping is unknown when μ and ξ are unspecified.

Estimation Methodology

Step 1: Estimating Shape Parameters

- For each firm i, we estimate by polynomial regression [Aït-Sahalia et al., 2021]:

$$\widehat{\boldsymbol{\Theta}}_{i,2} = \arg\min_{\boldsymbol{\theta}} \sum_{\kappa,\tau}\left(\sigma^{\mathrm{Market}}_{\mathrm{IV},i}(\kappa,\tau) - \left(\theta_{00} + \theta_{10}\kappa + \theta_{20}\kappa^2 + \theta_{01}\tau\right)\right)^2, \tag{6}$$

The estimated coefficients describe the shape of the IV surface of firm i.

Step 2: Estimating Credit Risk

- **For CDS spreads**. We estimate CDS in linear and nonlinear (quadratic) cases:

$$\begin{aligned} (\widehat{\alpha},\widehat{\boldsymbol{\beta}}) &= \arg\min_{\alpha,\boldsymbol{\beta}}\left\{\sum_i\left(\log \mathrm{CDS}_i - \alpha - \boldsymbol{\beta}^\top\widehat{\boldsymbol{\Theta}}_{i,2}\right)^2 + \lambda\|\boldsymbol{\beta}\|_1\right\}, \text{ or} \\ (\widehat{\alpha},\widehat{\boldsymbol{B}}) &= \arg\min_{\alpha,\boldsymbol{B}}\left\{\sum_i\left(\log \mathrm{CDS}_i - \alpha - \widehat{\boldsymbol{\Theta}}_{i,2}^\top\boldsymbol{B}\widehat{\boldsymbol{\Theta}}_{i,2}\right)^2 + \lambda\sum_{j,\ell}\left|B_{j\ell}\right|\right\} \end{aligned} \tag{7}$$

- **For credit rating**. We use a logit and latent threshold model. Let $R_i \in \{1,\ldots,K\}$ denote the credit rating of firm i, and the probability of each rating is

$$\begin{aligned} p_{ik} &= \mathbb{P}(R_i = k \mid \widehat{\boldsymbol{\Theta}}_{i,2}) = \frac{\exp\{g_k(\widehat{\boldsymbol{\Theta}}_{i,2})\}}{\sum_{\ell=1}^K \exp\{g_\ell(\widehat{\boldsymbol{\Theta}}_{i,2})\}}, \\ g_k(\boldsymbol{\Theta}) &= \alpha_k + \boldsymbol{\beta}_k^\top\boldsymbol{\Theta}, \quad \text{or} \quad g_k(\boldsymbol{\Theta}) = \alpha_k + \boldsymbol{\Theta}^\top\boldsymbol{B}_k\boldsymbol{\Theta}, \end{aligned} \tag{8}$$

We estimate the coefficients by ℓ_1-regularized MLE, where the loss is

$$-\sum_i \log p_{i,R_i} + \lambda\sum_{k=1}^{K-1}\|\boldsymbol{\beta}_k\|_1, \quad \text{or} \quad -\sum_i \log p_{i,R_i} + \lambda\sum_{k=1}^{K-1}\sum_{j,\ell}|B_{k,j\ell}|. \tag{9}$$

Empirical Results

- **Data**. We use daily equity option quotes, CDS spreads and S&P ratings for 328 firms from 1 January 2018 to 31 December 2025.
- **Out-of-sample results**. The nonlinear IV-surface model achieves the highest CDS R^2 and lowest RMSE, and outperforms all structural benchmarks across the three rating metrics.

	CDS spread			Credit rating		
	R^2	MAE	RMSE	Accuracy	MAE	Macro-F_1
• Panel A: Credit Information						
Rating/CDS	0.3851	0.4365	0.5444	0.5335	0.5091	0.4419
• Panel B: Benchmarks						
KMV	−1.5058	0.7414	1.0989	0.4451	0.6067	0.1910
Black–Cox	−2.0603	0.9511	1.2145	0.4451	0.6067	0.1910
CreditGrades	0.1352	**0.4137**	0.6456	0.4604	0.5854	0.2281
Heston	0.0360	0.5249	0.6816	0.4726	0.5732	0.2265
• Panel C: IV Surface						
Linear	0.3435	0.4417	0.5625	0.4604	0.6006	0.2243
Nonlinear	**0.4021**	0.4233	**0.5368**	**0.4970**	**0.5396**	**0.3348**

Conclusion

- We show that IV-surface shape parameters asymptotically identify risk-neutral default probabilities. A nonlinear mapping delivers informative out-of-sample predictions of CDS spreads and credit ratings.
- These results support that equity options can be used to infer credit risk information when ratings or CDS quotes are unavailable.

References

Y. Aït-Sahalia, C. Li, and C. X. Li. Implied stochastic volatility models. *The Review of Financial Studies*, 34(1):394–450, 2021.

P. Carr and L. Wu. A simple robust link between american puts and credit protection. *The Review of Financial Studies*, 24(2):473–505, 2011.

M. Cremers, J. Driessen, and P. Maenhout. Explaining the level of credit spreads: Option-implied jump risk premia in a firm value model. *The Review of Financial Studies*, 21(5):2209–2242, 2008.